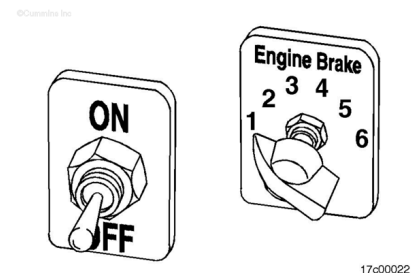


## Trainings ▾

### General Information

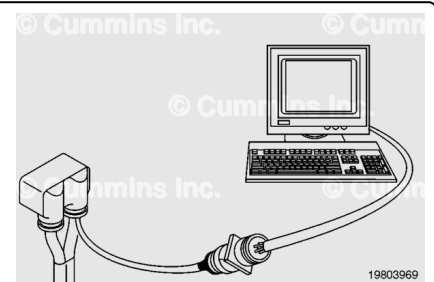
The engine brake ON/OFF switch circuit signals the system that the operator is requesting the engine brake system to be activated. The engine brake level switch determines what percentage of engine braking power will be used for engine braking. Three inputs to the electronic control module (ECM) from the engine brake level switch are used to communicate the setting to the ECM. Engine brake selector number 1 signal, engine brake selector number 2 signal, and engine brake selector number 3 signal in the original equipment manufacturer (OEM) connector are used. Various combinations of the three wires are used to represent the six switch positions.

After the ECM receives the signal from the engine brake ON/OFF switch and all other engine braking preconditions are met (engine RPM and road speed limits), the ECM will supply 12 VDC to the appropriate engine brake solenoids depending on how the 3 or 6 position engine brake level switch is set.



### Resistance Check

If an electronic service tool is available, monitor the engine brake ON/OFF switches for proper operation. If **not**, follow the troubleshooting procedures in this section.

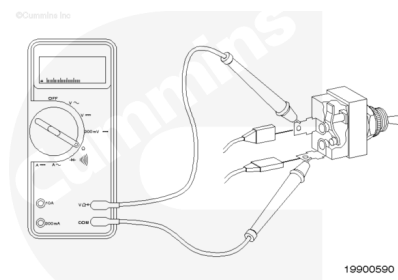


Locate the engine brake ON/OFF switch. Remove the electrical connectors from the switch.

Label the wires with the location on the switch or the wire number. Remove the electrical connectors from the switch.

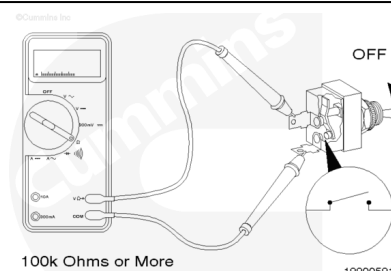
Adjust the multimeter to measure resistance.

Touch one multimeter probe to one of the terminals of the switch. Touch the other multimeter probe to the other terminal of the switch.



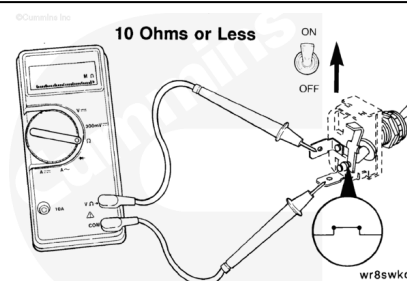
Move the switch to the OFF position. The multimeter **must** show an open circuit (100k ohms or more). If the circuit is **not** open, the switch has failed.

Refer to the OEM troubleshooting and repair manual for replacement procedures.



Place the switch in the ON position. The multimeter **must** show a closed circuit (10 ohms or less). If the circuit is **not** closed, the switch has failed.

Refer to the OEM troubleshooting and repair manual for replacement procedures.



## Check for Short Circuit to Ground

Touch one of the multimeter probes to one of the switch terminals. Touch the other probe to chassis ground. Move the switch to the ON position and measure the resistance. The multimeter **must** show an open circuit (100k ohms or more). If the circuit is **not** open, the switch has failed. Refer to the OEM troubleshooting and repair manual for replacement procedures. If the switch passes all of the previous checks, the circuit **must** be checked for an open circuit, a short circuit to ground, a short circuit from pin-to-pin, and a short circuit to an external voltage source.

